UNIT FINAL ASSESSMENT — TEACHER KEY & SCORING

> Original Grade 4 written final (26 items · 50 points · 30–45 min). Aligned to the official Unit Assessment blueprint (50 pts) — re-authored, not copied. Core Vocabulary deck may be used.

Total = 50 points

		Type		Expected answer	Reasoning / alternatives	Misconception targeted
1	1	select-all	2	**a, b, c, d** (all four)	Each shows a change caused by energy (motion, electricity, sound, chemical→motion/heat/light). Accept all four; ½ pt each.	Energy is invisible / "nothing happened" when only sound/light changed.
2	1	MC	2	**b — cause change**	Energy = ability to cause change. a/c/d are forms or effects, not the definition.	Confusing "energy" with a single form (heat/light).
3	1	diagram	4	Rubric below	Drawing shows	Change without

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					before→after change; energy type labeled; sentence describes change.	naming the energy form.
4	1	short	2	A change (sound/light/motion/heat) we observe is **evidence** energy was involved.	Any observable change counts; accept "we infer energy from change."	Equating observation with explanation.
5	2	MC	1	**c — energy of motion**	A moving object has energy of motion.	Calling it "stored energy."
6	2	MC	1	**b — a ball knocks a block over**	Moving object causes change. Others are stored/still.	Believing still objects have motion energy.
7	2	application	2	Any moving object causing an observable change, described.	Examples: ball → cup, wind → leaves, foot → ball. 2 pts = example +	Motion without a linked change.

		Type		Expected answer	Reasoning / alternatives	Misconception targeted
					described change.	
8	3	data table	2	Pattern: higher ramp → more cup movement → more energy of motion.	Accept "the higher the ramp, the farther the cup moved."	Confusing ramp height with the object's size/weight.
9	3	measured/ inferred	2	**Measure:** ramp height, cup movement. **Infer:** speed, energy of motion.	Accept "measure the cup; infer the speed."	Claiming we directly measure speed.
10	3	graph	2	As ramp height ↑, cup movement ↑ (line goes up). Evidence that faster (inferred) → more energy of motion.	Same-object rule applies.	Reading the axes backward.
11	3	short	3	For the SAME object, as speed increases, energy of motion increases (bigger change).	3 pts: relationship + same-object + change.	"Heavier = more energy" without control.
12	4	MC	1	**a — sound travels from a bell to your ears**	Sound transfers energy from place to place.	Sound is "nothing."
13	4	classify	1	**TRANSFER**	Warm water warms	Calling it transformatio

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					spoon; heat moves, form stays heat.	n.
14	4	select-all	2	**a, c, e** (bell heard, light on, backpack warm)	Evidence of transfer from place to place. ½ pt each.	"Light switch" as energy created.
15	4	classify	1	**TRANSFORMATION**	Battery stored energy → light: form changes.	Calling it transfer only.
16	4	diagram	2	INPUT: stored energy (battery) → OUTPUT: light (+ a little heat). Labels correct.	Accept "electrical → light."	Omitting the output.
17	5	MC	1	**b — energy of motion transfers to the target**	Collision: motion transfers; some → sound.	"Energy disappears."
18	5	data table	2	More weight → more blocks knocked → more energy of motion before collision.	Same release setup, change only weight.	Confusing weight with speed.
1	5	fair test	3	Fair test changes	3 pts: one-	Changing

		Type		Expected answer	Reasoning / alternatives	Misconception targeted
9				ONE thing (weight) and keeps everything else the same; predict more blocks fall.	variable + prediction + energy reasoning.	two things at once.
20	5	MC	1	**c — it does not disappear; it transfers (some becomes sound)**	Energy conserved; not gone.	"Energy disappeared."
21	6	classify	1	**BOTH**	Dominoes: motion transfers AND becomes sound (form changes).	Forcing either/or.
22	6	matching	3	criteria = what a solution must do · constraints = the limits · refine = improve from test evidence.	1 pt each.	Swapping criteria/constraints.
23	6	CER	3	Claim (device + what it does) · Evidence (what observed) · Reasoning (why evidence supports). Uses "supports," not "proves."	Rubric below.	Claim without evidence.
24	6	diagram	2	INPUT → TRANSFORMATION	Accept any valid safe	Reversing

		Type		Expected answer	Reasoning / alternatives	Misconception targeted
				→ OUTPUT chain labeled (e.g., electrical → motion → fan spins).	chain.	input/output.
25	7	MC	1	**b — he improved earlier designs through testing**	Edison did NOT invent the first bulb.	Lone-genius myth.
26	7	engineering application	3	Testing many materials (6,000+) + redesign (change one thing, test again) = persistence; improved the bulb.	3 pts: testing + redesign + persistence.	"He got it right first try" / lone genius.

Written-response rubrics

Item 3 — Draw a model of energy causing a change (4 pts)

- **4 — Above:** labeled before→after drawing; energy type named; sentence clearly describes the change.
- **3 — Average:** accurate drawing; energy type named; change described.
- **2 — Adequate:** drawing present; energy type OR description present.
- **1 — Inadequate:** no/inaccurate drawing; energy not named; change not described.

Item 23 — CER short explanation (3 pts)

- **3 — full:** claim + evidence + reasoning, linked; uses supports/shows.
- **2 — partial:** claim + evidence, reasoning weak/absent.
- **1 — minimal:** claim only.
- **0 — none/off-topic.**

Item 26 — Engineering application (3 pts)

- **3:** names testing (many materials/tries) + redesign (change one thing) + persistence.
- **2:** two of the three present.
- **1:** one present.
- **0:** off-topic or myth ("invented it alone").

What NOT to penalize

- Spelling/grammar (unless it changes the science).
- "Energy of motion" vs "motion energy" (both accepted).
- "Stored energy" vs "chemical energy" (project term = stored energy).
- Extra correct answers beyond the minimum.
- Using a Core Vocabulary card as reference (allowed).

Chapter/standard alignment (summary)

- Ch.1 → PE 4-PS3-2/4-PS3-3 context · Ch.2 → 4-PS3-1 · Ch.3 → 4-PS3-1 · Ch.4 → 4-PS3-2 · Ch.5 → 4-PS3-3 · Ch.6 → 4-PS3-4 · Ch.7 → ETS1.A + 4-PS3-4.